

Flat surface deflector

As per newton's second law of motion, the force of the jet strikes a plate is equal to the rate of change of momentum of the jet, therefore; For the flat surface deflector ($\theta = 90^\circ$):

$$F = \text{Mass of water flowing} \times \text{Change of velocity}$$

So,

$$F = \rho AV \times \Delta V = \rho AV \times (V \sin (90^\circ) - 0)$$

Therefore:

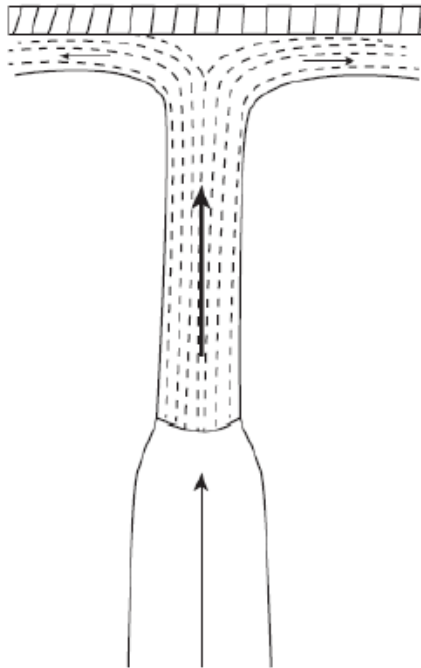
$$F = \rho AV^2$$

where:

$$V^2 = V_o^2 - 2gS$$

and,

$$V_o = Q/A$$



Where:

F = the force exerted by the jet in N.

ρ = density of water in kg/m³.

A = cross sectional area of the jet in m².

V = actual velocity at the center of the jet in m/s.

ΔV = change of the velocity in m/s.

V_0 = velocity of the jet in m/s.

S = the distant between the nozzle opening and the deflector opening in m.

g = is the gravitational acceleration in m/s².

Q = the flow rate in m³/s.

Answer:

$$A = 0.0000785 \text{ m}^2, S = 0.035 \text{ m}, g = 9.81 \text{ m/s}^2, \rho = 1000 \text{ kg/m}^3$$

$$\diamond Q = \text{-----} \text{ m}^3/\text{s}.$$

$$\diamond V_o = Q / A = \text{-----} = \text{-----} \text{ m/s}.$$

$$\diamond V^2 = V_o^2 - 2gS = \text{-----} = \text{-----} = \text{m/s}.$$

$$\diamond F_{th} = \rho A V^2 = \text{-----} = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$$

$$\diamond \text{Fac.} = \text{total weight} = \text{total mass} \times g = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$$

$\diamond$ Plot the graph between F_{th} & $F_{ac.}$ vs. Q :

Q (L/min.)									
F (theoretical)									
F (actual)									

